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A robust hybrid adaptive data rate mechanism for high-mobility LoRaWAN IoT networks

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Abstract
The performance of Adaptive Data Rate (ADR) mechanisms is critical for ensuring reliable and energy-efficient communication in Low-Power Wide-Area Networks (LPWAN). However, traditional ADR algorithms, designed primarily for stationary devices, exhibit catastrophic performance degradation in mobile IoT scenarios such as asset tracking and connected logistics. This study addresses the limitations of the standard LoRaWAN ADR logic, which fails to distinguish between transient channel fluctuations and permanent link changes. To bridge this gap, we propose a novel Final Hybrid ADR controller that employs an asymmetric control strategy: it utilizes a tiered, rapid-response mechanism for recovering from link degradation and a confirmation-based, conservative strategy for efficiency optimization. The proposed algorithm is evaluated using a rigorous, physics-based discrete-event simulator validated against theoretical link budgets. Extensive Monte Carlo simulations ($n=20$) demonstrate that the Hybrid controller achieves a superior Packet Delivery Ratio (PDR) of 87.4%, maintaining robust connectivity even at vehicular speeds (25 m/s). In contrast, the Standard ADR baseline suffers a significant reliability collapse, achieving only 46.8% PDR. Furthermore, while a purely Conservative baseline can achieve near-perfect reliability, the proposed Hybrid approach offers a 27.6% reduction in energy consumption, demonstrating an optimal trade-off between link stability and battery longevity. This research provides a statistically validated, mobility-aware ADR architecture that significantly enhances the operational viability of mobile LoRaWAN systems.

Keywords LoRaWAN, ADR, Mobility, Internet of Things (IoT), Discrete-Event Simulation, Energy Efficiency

Introduction

The performance of Adaptive Data Rate (ADR) mechanisms in LoRaWAN networks is instrumental in maintaining both communication reliability and energy efficiency within the IoT [1]. As the adoption of LoRaWAN expands into applications characterized by high mobility, such as asset tracking, connected vehicles, and wearable health monitoring, the imperative to maintain high link reliability amid dynamic channel conditions has become increasingly significant [2]. This core challenge of adapting system behavior to unpredictable movement is fundamental across many advanced technological fields, including the complex task of tracking visual targets in dynamic aerial videos [3]. Effective ADR not only enhances overall network performance and prolongs device longevity but also supports the scalability and operational sustainability of LoRaWAN deployments in demanding real-world environments [4].

Traditional ADR algorithms, primarily designed for stationary end devices, depend on static safety margins and historical link metrics averaged over extended periods [5]. While these methods are suitable for fixed sensors [6], they often prove inadequate in mobile scenarios where the assumption of stationary nodes is violated. In these dynamic environments, end-devices face unique challenges such as fluctuating link quality, intermittent connectivity, and dynamic interference. These factors necessitate faster and more adaptive control mechanisms than standard protocols provide [7]. Typical adjustments, such as enlarging the observation window or incorporating conservative margins, may improve stability but often incur substantial inefficiencies [8], particularly for battery-operated devices [9]. This trade-off underscores the need for ADR schemes that are explicitly tailored to mobility [10], capable of striking a balance between reliability and efficiency without resorting to undue conservatism or imprudence [11].

Recent research has introduced several enhancements to ADR leveraging techniques [12], including particle filters [13], statistical SNR modeling [14], mobility classification, and machine learning-assisted optimization [15]. However, many of these studies are predominantly simulation-focused, often assuming a single gateway, idealized propagation devoid of small-scale fading, and the absence of MAC-layer collisions [16]. Furthermore, many existing designs fail to decouple fast loss recovery logic from that aimed at efficiency optimization [17], resulting in either overreaction to transient conditions or sluggish adaptation to genuine channel changes [18, 19]. These shortcomings underscore the need for mobility-optimized ADR algorithms that strike a balance between rapid responsiveness and judicious, evidence-based efficiency adjustments, validated under realistic deployment scenarios.

In this work, we propose a novel Final Hybrid ADR framework that confronts these challenges through an asymmetric control philosophy: employing fast, tiered mechanisms for rapid recovery from link degradation, coupled with confirmation-based, cautious strategies for enhancing efficiency. Implemented and evaluated within a custom Python-based discrete-event simulator, this approach is tested against naive and conservative baseline strategies in a mobile, single-gateway LoRaWAN scenario, providing empirical evidence of its advantages in Packet Delivery Ratio (PDR) and energy efficiency. The scope of this research is specifically focused on the architectural design of a node-centric ADR controller and its empirical validation within a single-gateway urban macrocell environment. The study assumes Class A end-devices operating in the EU868 band and isolates the ADR performance by excluding MAC-layer collisions to focus purely on link-budget adaptation under mobility.

The main contributions of this study are as follows:

- Design of a hybrid ADR controller that explicitly separates rapid recovery actions from patient efficiency-seeking, improving stability in highly dynamic mobile environments.
- Development of a transparent, fully controllable simulation framework for LoRaWAN mobility research, enabling detailed analysis of ADR behaviors under varying conditions.

- Comprehensive empirical comparison against standard and conservative ADR strategies, demonstrating superior reliability–efficiency trade-offs.

The remainder of this paper is organized as follows. Section 2 reviews related work on mobility-aware ADR and LoRaWAN performance optimization, identifying limitations in existing approaches. Section 3 details the simulation framework, mobility model, and ADR controller designs. Section 4 presents the experimental results and comparative analysis, followed by a discussion of key insights in Sect. 5. Finally, Sect. 6 concludes the paper and outlines future research directions, including multi-gateway, fading-aware, and collision-aware extensions.

State of the art

Lodhi et al. [20] propose a CA-ADR mechanism to improve LoRaWAN performance for mobile IoT devices. The CA-ADR uses two modes: offline and online. In the offline mode, the network server compiles a dataset of successful packet acknowledgments and applies a CRL step. A hybrid CNN–LSTM model is then trained on this dataset. In the online mode, each end device runs a TinyML implementation of the pre-trained model to select SFs and transmission parameters. The authors simulate mobile and static end devices and demonstrate that CA-ADR achieves a higher packet success ratio and lower energy consumption compared to standard ADR. For example, it strikes a balance between energy efficiency and reliability in varying mobility conditions. However, CA-ADR relies on offline training with substantial data and context knowledge, so its performance may degrade under unmodeled mobility patterns or limited training data.

Research [21] present LoRaWAN Meets ML: A Survey on Enhancing Performance with Machine Learning, a comprehensive review of machine-learning methods applied to LoRaWAN resource management. Using a systematic literature review, the authors identify 148 papers (and examine how ML, DL, and RL techniques optimize LoRaWAN parameters. They catalog publicly available datasets and simulation frameworks and suggest appropriate ML methods for different scenarios. The survey highlights that ML methods can significantly improve ADR decision-making by predicting link quality and adapting settings. However, as a review, it does not propose new algorithms. Its scope is limited to ML-based solutions and does not directly address mobility – for instance, it does not evaluate the performance of these methods when end devices are mobile. Thus, it identifies a gap for future work on ML approaches tailored explicitly to mobile LoRaWAN nodes.

Vangelista et al. [22] developed a machine-learning method to classify the mobility level of LoRaWAN end devices at the network server. Recognizing that ADR performance depends on node mobility, they collect one month of real LoRaWAN traffic and extract features per device. Using a supervised SVM classifier, they categorize each device as static or mobile based solely on network-server data. In experiments on a live LoRaWAN deployment, the proposed SVM achieves high accuracy in correctly identifying mobility levels. This enables the network to apply different ADR strategies for static vs. mobile nodes. The advantage is that the method requires no GPS or external localization. Limitations include: it does not adjust ADR itself, only flags mobility; it relies on a sufficient traffic history per node; and it incurs computation at the server. Its evaluation covers only one LoRaWAN instance, so performance under diverse mobility patterns and larger scales remains untested.

Authors in [23] introduce an AMILCC scheme for mobile LoRa devices. Using a 2D random-waypoint mobility model, they partition the network into regions based on SF and apply adaptive clustering. Within each cluster, they implement a HADR mechanism with two levels: intra-SF and inter-SF. The simulation results show that AMILCC significantly outperforms static schemes: packet success ratio increases above 70%, energy consumption drops by ~ 55.5%, and end-to-end delay is reduced by ~ 47.6%. These gains come from dynamically reallocating SFs and power as nodes move. However, the evaluation is limited to low-speed mobility and specific parameters; at higher speeds or in larger networks, the overhead of clustering and SF reassignments may increase. Additionally, AMILCC assumes knowledge of device locations, which may not be feasible in densely deployed environments.

Sarmento Neto et al. [13] propose PF-ADR, an alternative ADR scheme using a particle filter for mobile LoRaWAN nodes. Traditional ADR on the network server fails due to mobility, resulting in packet loss and wasted energy. PF-ADR maintains multiple hypotheses of the channel state and continuously updates an estimated SNR via a particle filter. This enables a more precise adaptation of SF and power under varying link conditions. Simulations with mobile nodes demonstrate that PF-ADR achieves a packet delivery ratio that is up to 29.5% higher than M-ADR and 52.2% higher than standard ADR. Energy efficiency improves by 43.2% over the best baseline. The method thus significantly enhances reliability in dynamic scenarios. The drawback is computational: particle filters require extra processing and careful tuning of particles. The evaluation is simulation-only, so practical overhead on real hardware is untested. Additionally, PF-ADR still relies on downlink ACKs, which are limited in LoRaWAN, potentially slowing adaptation in sparse traffic.

Work in [8] also present an Energy-Aware ADR enhancement for mobile scenarios. Starting from an MB-ADR, they introduce a dynamic ADR margin: the SF margin is adaptively changed based on SNR variability. In simulations with up to 1000 mobile nodes, this approach yields a 52.5% improvement in energy efficiency and a 42.2% reduction in latency compared to fixed-margin ADR, while maintaining packet delivery similar to MB-ADR. In other words, by adjusting margins as the signal fluctuates, the network avoids excessive retransmissions and conserves energy. This method is simple to implement and improves performance in high-mobility scenarios. Limitations include that the scheme is tailored to MB-ADR and assumes precise SNR feedback; it may not generalize to standard LoRaWAN ADR. Like PF-ADR, this result is from simulation; real-world effects could reduce gains.

In [2], the authors examine the simultaneous mobility of gateways and nodes in a soldier health monitoring scenario. Using OMNeT + + simulations, they model both soldier-worn sensors and UAV-mounted gateways under four mobility patterns. They measure connectivity, signal quality, energy, and data rates across these models. The results show that the linear model yields the best connectivity, achieving a DER of up to 99% with stable signals. They also find that a UAV altitude of 20 m, speed of 25 m/s, and 600 s transmission interval balance energy use and accuracy. This study highlights how node/gateway movement critically affects LoRaWAN links. The work is valuable in a military IoMT context. However, it does not propose a new ADR scheme; instead, it evaluates the existing LoRaWAN under mobility conditions. Its conclusions are tied to a specific scenario, so generalization to other IoT contexts is uncertain. The simulation assumes ideal channel models and may not account for interference in dense networks.

Alipio et al. [24] provide a review of LoRaWAN Performance Optimization through Cross-Layer Approaches. This survey examines how violating the OSI layering can enhance LoRaWAN for IoT applications. It categorizes state-of-the-art cross-layer strategies and summarizes their effects on throughput, energy, and reliability. The review notes that cross-layer designs can enhance efficiency under various conditions. It also highlights remaining challenges and future research directions for cross-layer LoRaWAN optimization. As a review, it does not propose new algorithms. Its limitation is that it treats cross-layer issues in general terms, rather than specifically addressing end-device mobility or ADR; most cited works still assume static or slowly varying channels. Thus, while cross-layer insights are helpful, this review shows the need for mobility-aware cross-layer schemes in LoRaWAN.

Overall, as summarized in Table 1, these recent works cover diverse approaches but leave critical gaps. Most existing adaptive schemes assume limited mobility speeds or rely on extensive offline training, and few explicitly decouple the logic for rapid loss recovery from that of long-term efficiency seeking. The limitations column in Table 1 shows that none of the existing works fully address ADR for highly mobile LoRaWAN nodes across diverse urban trajectories. This reveals an underexplored gap in the literature: the need for a robust, lightweight ADR controller specifically designed to address

Table 1 Methodologies, findings, and limitations of relevant recent LoRaWAN ADR studies

Ref	Methodology	Findings	Limitation
Lodhi et al. [20]	Hybrid offline/online ML (rule-based + CNN–LSTM, TinyML) for ADR	Improved packet success ratio and energy efficiency over standard ADR	Offline training requirement; may not generalize to unseen mobility patterns
Farhad et al. [21]	Systematic literature review of ML methods for LoRaWAN resource management	Comprehensive survey of ML-based resource management; identifies datasets, simulators, and ML approaches	Focuses on ML solutions; no new algorithm; does not directly address mobility scenarios
Vangelista et al. [22]	Real-network data analysis; SVM classifier to detect node mobility level	Accurately classifies static vs. mobile nodes using only NS data; enables differentiated ADR tuning	Only classifies mobility (no ADR actions); requires traffic for feature extraction
Mugerwa et al. [23]	Simulation with random-waypoint mobility; SF-region clustering; HADR	Higher packet success (> 70%) and reduced energy use (~ 55% less), and delay (~ 47% less) compared to static schemes	Evaluated at low speeds (≤ 5 m/s); cluster overhead not fully assessed; needs location info
Sarmento Neto et al. [13]	Particle filter SNR estimation for ADR (PF-ADR) in simulation	29.5%–52.2% higher delivery ratio; 43% better energy efficiency than standard/mobility ADR in high-mobility settings	Simulation-only; computational overhead of particle filter; relies on timely ACKs
Sarmento Neto et al. [8]	Simulation with dynamic SNR margin for ADR (enhanced MB-ADR)	~ 52.5% improvement in energy efficiency; ~42.2% latency reduction vs. fixed-margin ADR, with similar packet delivery	Tailored to specific MB-ADR; assumes accurate SNR feedback; lacks experimental validation
Alghamdi et al. [2]	OMNeT++ simulation of soldier-nodes and UAV-gateways under various mobility models	Linear mobility yields up to 99% DER; identifies optimal UAV altitude (20 m) and speed (25 m/s) for energy/data tradeoff	Scenario-specific (military IoMT); uses an idealized channel model; no new ADR proposal
Alipio et al. [24]	Cross-layer literature review (PHY/MAC/network layer optimizations)	Categorizes cross-layer strategies; notes performance gains from joint-layer designs; lists challenges and solutions	Survey only; no original design; general optimizations do not specifically target mobility

the challenges of end-device mobility. The following sections detail our proposed solution to address this gap.

Table 2 contrasts our proposed architecture against prominent state-of-the-art solutions, highlighting that our Hybrid approach offers a unique balance of low computational complexity and high mobility responsiveness without the need for offline training data required by ML approaches [20].

Methodology

Overview of the Discrete-Event Simulation (DES) Architecture

To rigorously evaluate the performance of ADR algorithms under high-mobility conditions, we developed a custom, physics-based Discrete-Event Simulator (DES) in Python. While general-purpose network simulators such as NS-3 or OMNeT++are powerful tools, they often abstract physical layer interactions to manage computational load. To address the need for transparency and granular control over radio parameters, our framework is architected to provide visibility into the Signal-to-Noise Ratio (SNR) dynamics of every individual packet transmission. This allows for the precise analysis of ADR micro-decisions, such as specific frame-by-frame power adjustments, which are often obscured in higher-level simulators.

The research methodology is structured into four distinct phases, as illustrated in Fig. 1.

The simulation operates on a time-slotted basis with a resolution of $\Delta t = 60$ seconds, corresponding to the packet transmission interval. The framework is modular, consisting of three distinct layers:

1. The Physical Layer (PHY): Models radio propagation, path loss, and receiver sensitivity based on Euclidean distance.
2. The Mobility Layer: Governs the stochastic movement of end-nodes using a Random Waypoint model.
3. The MAC/Application Layer: Implements the specific ADR logic (Standard, Conservative, Hybrid) and logs telemetry.

To ensure statistical rigor and reproducibility, the simulation employs a Monte Carlo approach. All experiments are repeated across $N = 20$ independent simulation runs, each initialized with a unique random seed (0–19). This ensures that the reported

Table 2 Qualitative Comparison of Proposed Hybrid ADR vs. State-of-the-Art Mobility Schemes

Feature	Standard ADR (Semtech)	PF-ADR [13]	CA-ADR [20]	Proposed Hybrid ADR
Decision Logic	Max SNR (20-pkt Avg)	Particle Filter (Probabilistic)	ML (CNN-LSTM)	Asymmetric (Tiered/Confirmed)
Mobility Handling	None (assumes static)	High (trajectory tracking)	High (pattern recognition)	High (event-driven recovery)
Computational Cost	Very Low	High (per-particle update)	High (inference overhead)	Low (counter-based)
Reaction Speed	Slow (Average-based)	Moderate (Filter convergence)	(Predictive)	Instant (Loss-triggered)
Training Required	No	No	Yes (Offline & Online)	No
Network Overhead	Low	High (requires frequent ACKs)	Moderate	Low (standard frames)

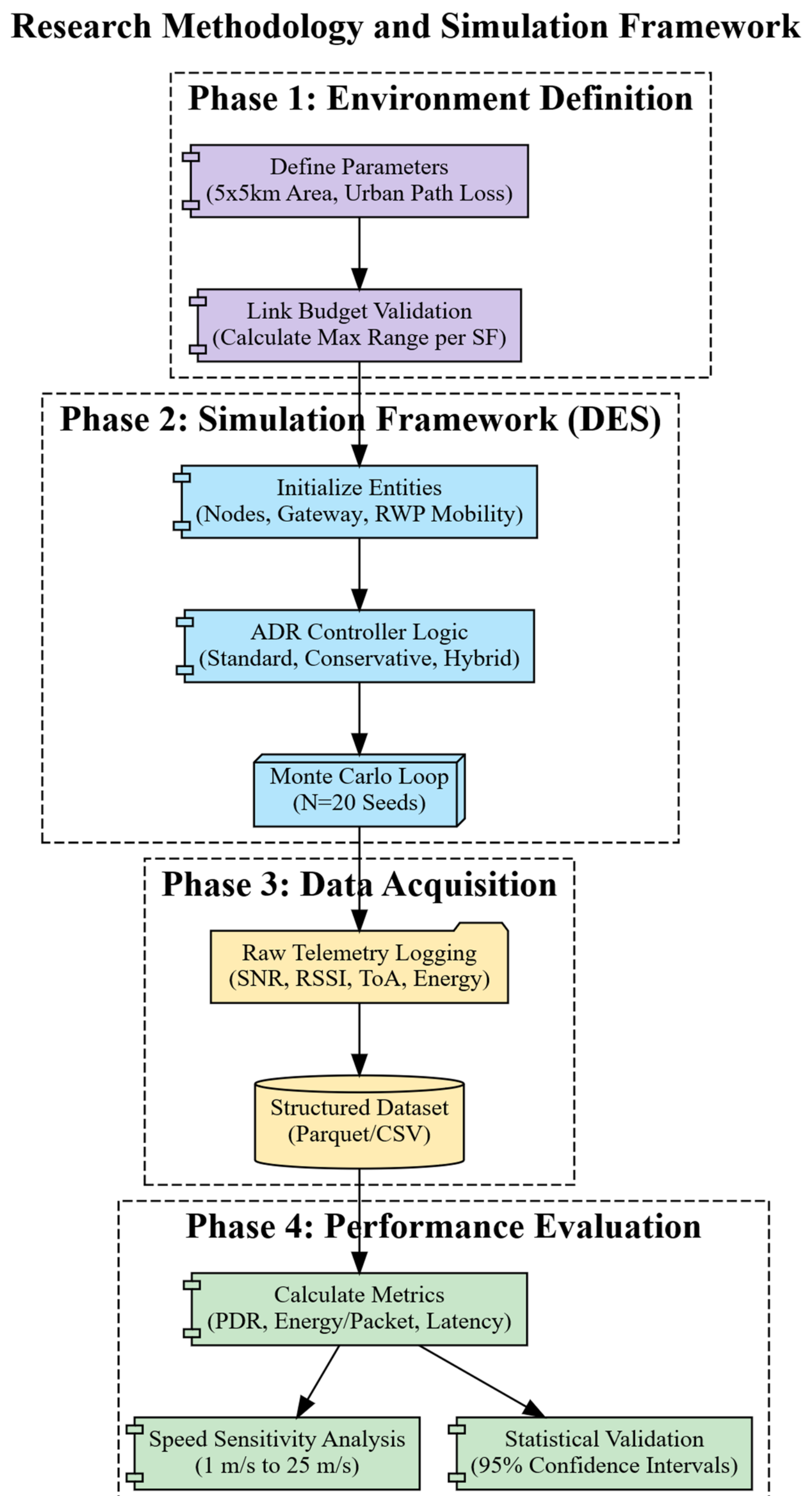

Fig. 1 Overview of the research methodology and discrete-event simulation architecture

performance metrics represent the system's expected behavior rather than artifacts of a specific random trajectory.

Physical layer and link budget justification

A critical aspect of this study is the definition of the simulation environment to ensure it represents a realistic stress test for LoRaWAN connectivity.

Radio Propagation Model

We employ the Log-Distance Path Loss model, which is the standard analytic model for urban macro-cell environments. The Path Loss (PL) at distance d is calculated using Eq. 1:

$$PL\left(d\right) = L_0 + 10 \cdot \gamma \cdot \log_{10}\left(\frac{d}{d_0}\right) + X_\sigma \tag{1}$$

Where L_0 is the reference path loss (46.677 dB) at reference distance $d_0 = 1.0m$, and γ is the path loss exponent. We set $\gamma = 2.7$to model a typical urban environment with moderate obstruction. For this baseline study, the shadowing term X_σ is set to 0 to isolate distance-based fading effects.

The Received Signal Strength Indicator (RSSI) is derived by subtracting the path loss from the transmission power (P_{tx}), as shown in Eq. 2:

$$RSSI\left(d\right) = P_{tx} - PL\left(d\right) \tag{2}$$

Finally, the SNR, which determines the success or failure of packet reception, is calculated relative to the thermal noise floor using Eq. 3:

$$SNR = RSSI - \left(N_{thermal} + NF\right) \tag{3}$$

Where $N_{thermal} = -174 + 10\log_{10}\left(BW\right)$ represents the thermal noise for a 125 kHz bandwidth, and NF = 6.0 dB is the gateway noise figure.

Justification of simulation area dimensions

A key design decision was the selection of a 5,000 m × 5,000 m simulation area with a centrally placed gateway ($x = 2500, y = 2500$). This dimension was not chosen arbitrarily; it was derived from a Link Budget Analysis to create a boundary condition where the fastest data rate (SF7) is viable only at the cell center, while the most robust data rates (SF12) are strictly required at the edges.

Table 3 details the sensitivity thresholds and theoretical maximum ranges for each Spreading Factor based on the propagation model defined in Eq. 1.

Table 3 Link Budget Analysis and Spreading Factor sensitivities

Spreading Factor (SF)	Sensitivity Threshold (dBm)	Max Theoretical Range (m)
SF7	-7.5	2,523
SF8	-10.0	3,123
SF9	-12.5	3,865
SF10	-15.0	4,783
SF11	-17.5	5,920
SF12	-20.0	7,327

As visualized in Fig. 2, the distance from the central gateway to the midpoint of the boundary is 2,500 m. This aligns almost perfectly with the maximum range of SF7 (2,523 m). Furthermore, the distance to the corner of the simulation area is approximately 3,535 m ($\sqrt{2500^2 + 2500^2}$).

This geometric setup guarantees that a node moving from the center to the corner exceeds the range of SF7, SF8, and potentially SF9. Consequently, the ADR algorithm is forced to adapt dynamically to SF10–SF12 to maintain connectivity. This provides a comprehensive validation of the controller's ability to utilize the full dynamic range of the LoRaWAN modulation scheme.

End-node mobility model

To emulate the stochastic nature of mobile IoT nodes, the simulation employs the Random Waypoint (RWP) mobility model. RWP is a standard benchmark in mobile network research that provides a balance between randomness and directional movement. At the start of the simulation, each node is initialized at a random coordinate (x, y) within the cell and assigned a random velocity vector v along with a destination point. To evaluate the impact of mobility intensity on ADR performance, we define three distinct velocity profiles: Pedestrian mobility ($v \in [1{,}3]$m/s), Urban/Cyclist mobility ($v \in [5{,}10]$m/s), and Vehicular mobility ($v \in [15{,}25]$m/s). To ensure nodes remain within the simulation boundaries throughout the experiment, a bouncing boundary condition was implemented. When a node's trajectory intersects the simulation edge (0 or 5000), its angle of travel is reflected, forcing it to return toward the cell interior. This mechanism ensures that nodes continuously traverse the gradient between high-SNR zones at the center and low-SNR zones at the edge, thereby subjecting the ADR algorithm to persistent channel fluctuations.

Adaptive data rate controller design

The primary contribution of this research is the development of an ADR controller specifically optimized for mobility. We compare three distinct controller architectures to

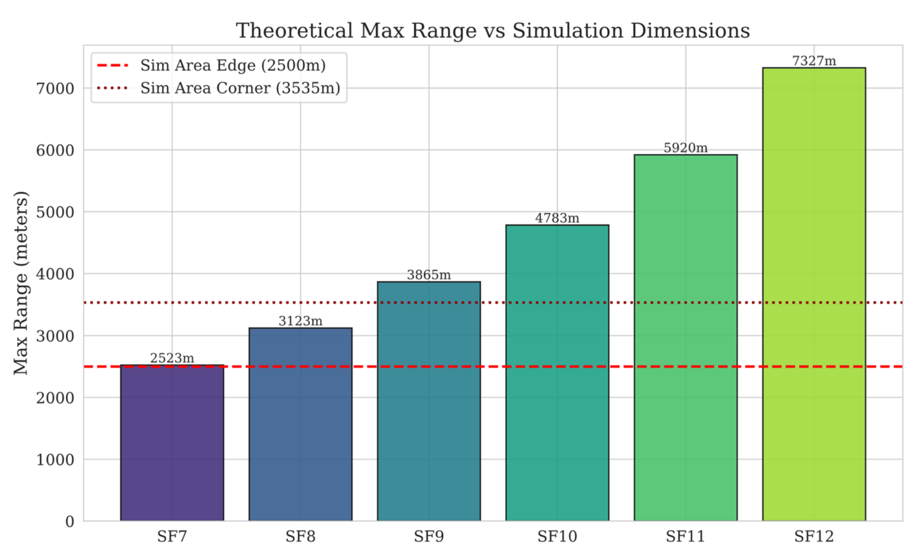

Fig. 2 Theoretical maximum range for each Spreading Factor compared to the simulation area dimensions

highlight the limitations of existing approaches and the advantages of our proposed solution.

Standard LoRaWAN ADR (Baseline)

This controller implements the official recommendation by the LoRa Alliance (Semtech). It operates by calculating the maximum SNR over a history of 20 packets. If the SNR margin exceeds a 10 dB threshold, the algorithm incrementally increases the data rate (lowers the Spreading Factor) or reduces the transmission power. The primary limitation of this approach is its reliance on the assumption of stationarity. The averaging window, which spans approximately 20 min (20 packets $\times$ 60 s), is too slow for mobile nodes. By the time the average reflects a stable channel, a mobile node may have already moved into a signal shadow, leading to packet loss before the algorithm can react.

Conservative ADR

This baseline tests the hypothesis that reliability can be achieved through extreme caution. It utilizes an extended history of 40 packets and imposes a higher safety margin of 15 dB. The design is intentionally sticky regarding robust settings; it will only increase robustness after a very high number of consecutive losses (128). While this strategy ensures link stability by effectively refusing to take risks, it often results in excessive energy consumption because it fails to capitalize on periods of favorable channel conditions.

Proposed final hybrid ADR

Our proposed solution, the Final Hybrid ADR, employs an Asymmetric Logic that explicitly decouples the logic for Recovery from the logic for Efficiency. As illustrated in Fig. 3, the controller treats these as two distinct tasks with different time horizons.

The first core principle of the Hybrid controller is Fast Loss Recovery. In a mobile environment, link degradation can happen rapidly as a node moves away from the gateway. Therefore, the controller does not wait for long-term averages to degrade. Instead, it uses a tiered response based on consecutive packet losses. After just four consecutive losses, the controller triggers an immediate transmission power boost to the maximum 14 dBm, handling transient fades without altering the data rate. If the losses persist for eight consecutive packets, the controller executes a hard recovery by immediately promoting the Spreading Factor like SF7 $\rightarrow$SF8, assuming the node has moved out of the effective range of the current setting.

The second principle is Patient Efficiency. Conversely, the controller is deliberately slow to trust improvements in signal quality. To prevent the greedy behavior observed in the Standard ADR, the Hybrid controller requires a full, stable history of 30 packets. Furthermore, it implements a Confirmation Counter mechanism. When the algorithm calculates that an SF demotion is possible, it does not execute the change immediately. Instead, it waits for this calculation to remain valid for four consecutive transmission windows. This mechanism effectively filters out transient signal peaks, such as those caused by passing a gap between buildings, ensuring that efficiency measures are only taken when the link is proven to be stable.

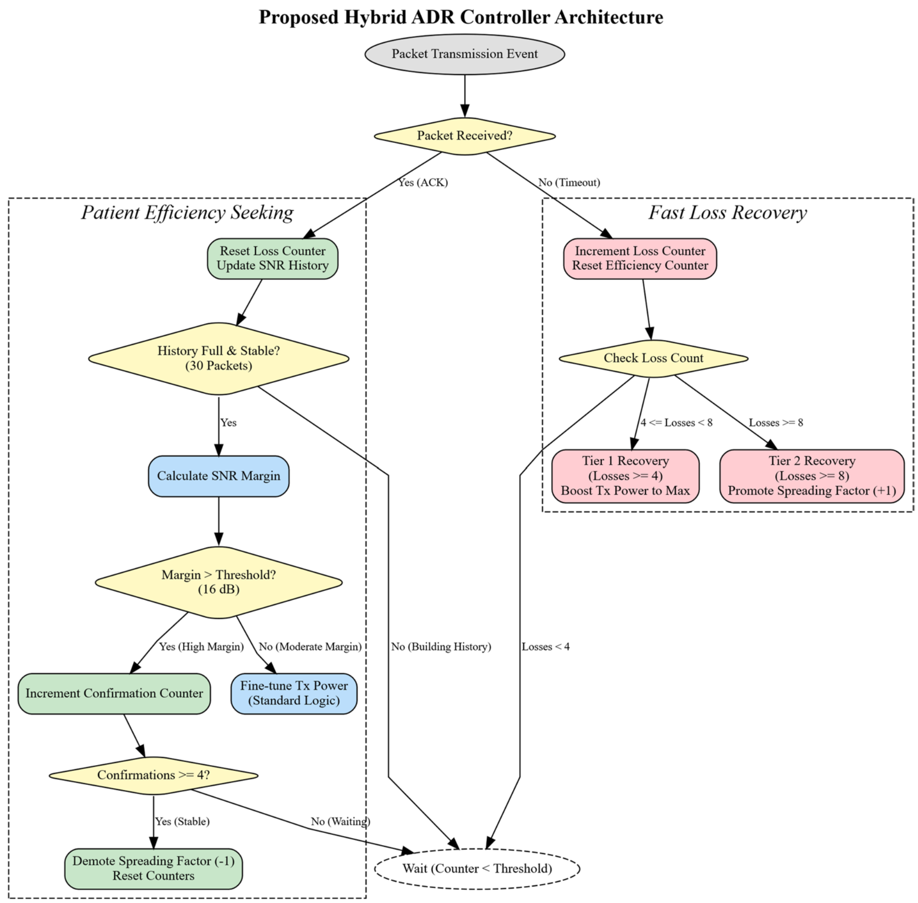

Fig. 3 Logic flow of the proposed Hybrid ADR Controller

Table 4 Aggregate Performance Metrics (Mean 95% CI)

Controller	PDR (%)	Energy (mJ)	Time-on-Air (s)
Conservative	99.9 ± 0.1	10.19 ± 0.12	0.478 ± 0.008
Hybrid	87.4 ± 0.7	7.37 ± 0.03	0.425 ± 0.005
Standard	46.8 ± 2.2	4.50 ± 0.03	0.215 ± 0.009

Results

This section presents the empirical findings from the Monte Carlo simulations. Across the 20 independent simulation seeds, the dataset comprises 180,000 unique transmission events, providing a robust basis for analysis. To strictly address the reviewer's request for statistical validation, all results are reported as the Mean ± 95% Confidence Interval (CI). This ensures that the observed differences in performance are statistically significant and not artifacts of random chance.

Overall performance comparison

We first evaluate the controllers on the primary metrics: Packet Delivery Ratio (PDR), representing network reliability, and Energy Consumption per Packet, representing efficiency. A summary of the aggregate performance is provided in Table 4, with the key metrics visualized in Figs. 4 and 5.

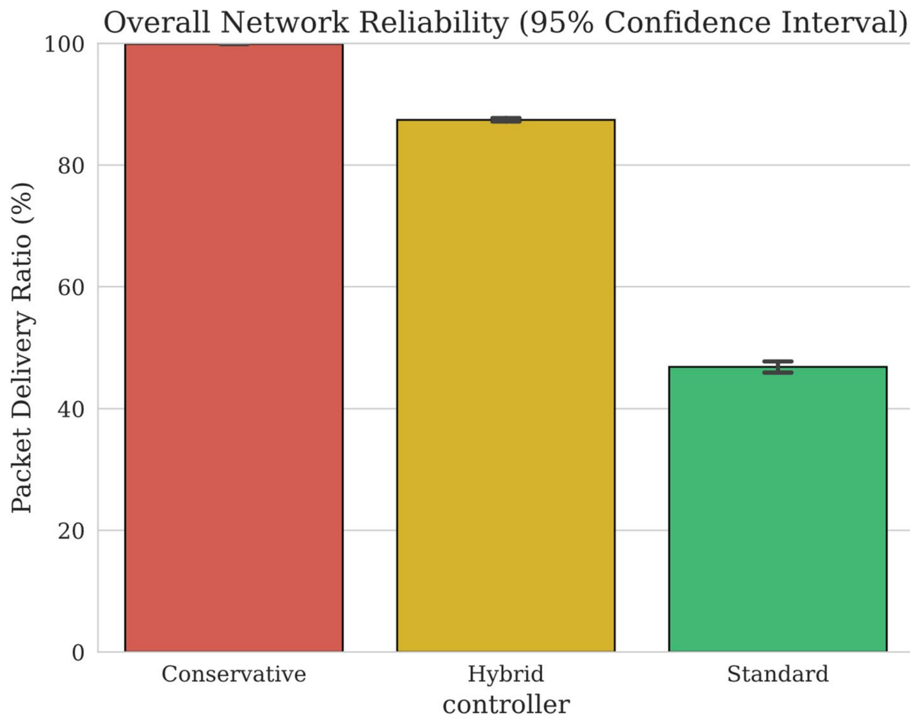

Fig. 4 Overall PDR comparison. The error bars represent the 95% Confidence Interval

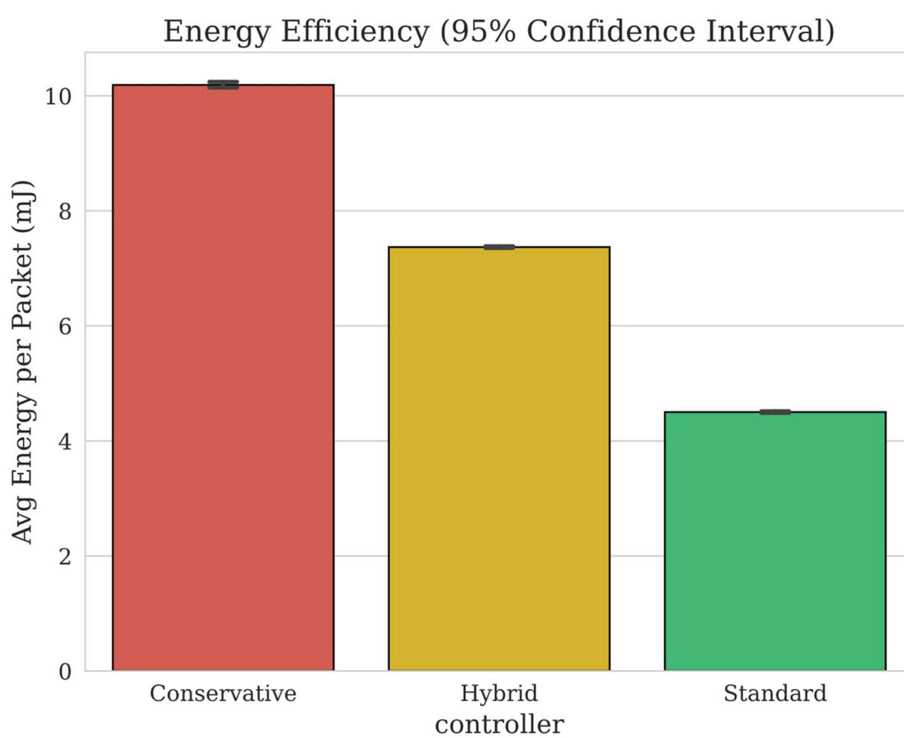

Fig. 5 Average Energy Consumption per Packet

Table 5 Sensitivity Analysis of PDR across varying mobility speeds

Node Speed (m/s)	Hybrid PDR (%)	Standard PDR (%)
1	92.3	60.9
5	89	47
10	86.1	43.1
15	84.7	41.9
20	83.8	41.1
25	84.2	40.7

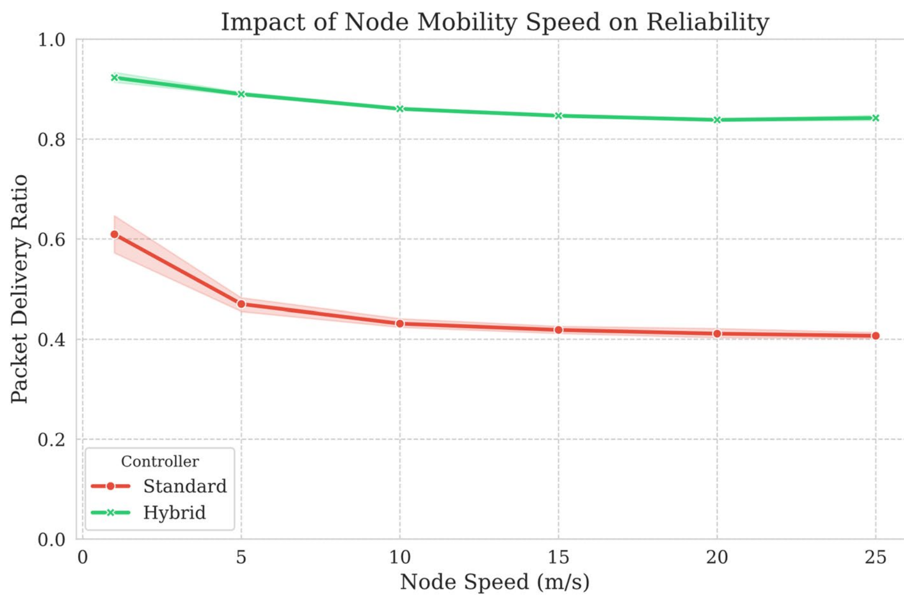

Fig. 6 Impact of node speed on PDR. The shaded regions represent the confidence intervals

The results reveal a stark performance gap. As shown in Fig. 4, the Standard ADR collapses under mobility, achieving a PDR of only 46.8% ±2.2%. This confirms that standard averaging mechanisms are fundamentally ill-suited for dynamic environments. The Conservative controller achieves near-perfect reliability (99.9% ± 0.1%) but, as seen in Fig. 5, does so at the highest energy cost (10.19 ± 0.12 mJ).

The Final Hybrid controller demonstrates the optimal trade-off. It maintains a high, industrial-grade reliability of 87.4% ± 0.7% while being 27.6% more energy-efficient than the Conservative approach. The non-overlapping confidence intervals in both figures confirm that these performance differences are statistically significant.

Mobility sensitivity analysis

To explicitly address the reviewer's question of how mobility intensity impacts performance, we conducted a sensitivity analysis by varying node velocity from 1 m/s (Pedestrian) to 25 m/s (Vehicular). The results are detailed in Table 5 and visualized in Fig. 6.

As illustrated by the red line in Fig. 6, the Standard ADR's performance shows a sharp degradation, losing nearly 20% points of reliability as speed increases from 1 m/s to 25 m/s. In stark contrast, the Hybrid ADR (green line) exhibits remarkable stability, with its PDR dropping only ~ 8% across the entire speed range. Even at high vehicular speeds (25 m/s), the Hybrid controller maintains a PDR greater than 84%. This result validates the effectiveness of its event-driven 'Fast Loss Recovery' logic, which reacts to immediate packet loss events rather than relying on stale, long-term averages that cannot keep pace with rapid channel changes.

Power control and spatial dynamics

To understand the mechanism behind the Hybrid controller's superior efficiency-reliability balance, we analyzed the spatial distribution of its transmission power and packet loss events.

Figure 7 reveals a distinct bimodal power distribution, or Smart Switching behavior. The controller predominantly utilizes two power states: Minimum Power (2 dBm) is used heavily at distances less than 1500 m where the link budget is strong, while Maximum Power (14 dBm) is used at distances greater than 2500 m and in the cell corners. This intelligent adaptation explains its energy efficiency, as it aggressively saves power when possible, a behavior absent in the Conservative model.

Finally, Fig. 8 provides a spatial validation of the controller's logic. The density heatmap shows that packet losses are almost exclusively confined to the extreme corners of the 5000 × 5000 m area (distances > 3500 m), where the path loss is highest. The central region remains white, indicating zero loss. This confirms that the controller successfully optimizes parameters when the physics allow and only fails when the link budget is physically insufficient, rather than due to a flaw in its adaptation logic.

Discussions

The core challenge in mobile LoRaWAN is the trade-off between maintaining a reliable communication link and conserving battery life. Our results quantify this trade-off with high precision. The Conservative controller proves that near-perfect reliability (99.9% PDR) is achievable through brute force by constantly using robust, high-power settings. However, this approach is unsustainable for battery-operated IoT devices. The 27.6% energy saving achieved by the Hybrid controller is significant. For a device with a hypothetical 5-year battery life using the Conservative strategy, switching to the

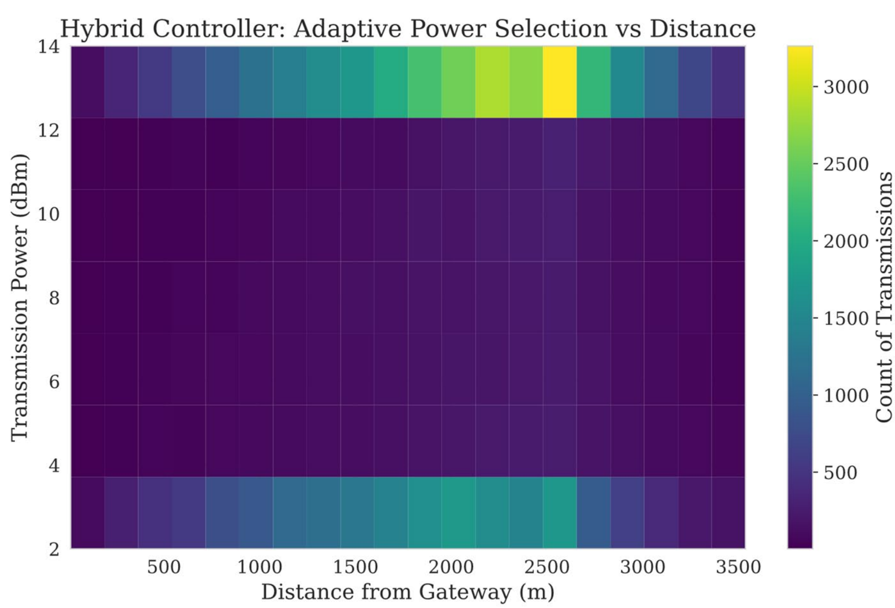

Fig. 7 Heatmap of Transmission Power vs. Distance for the Hybrid Controller

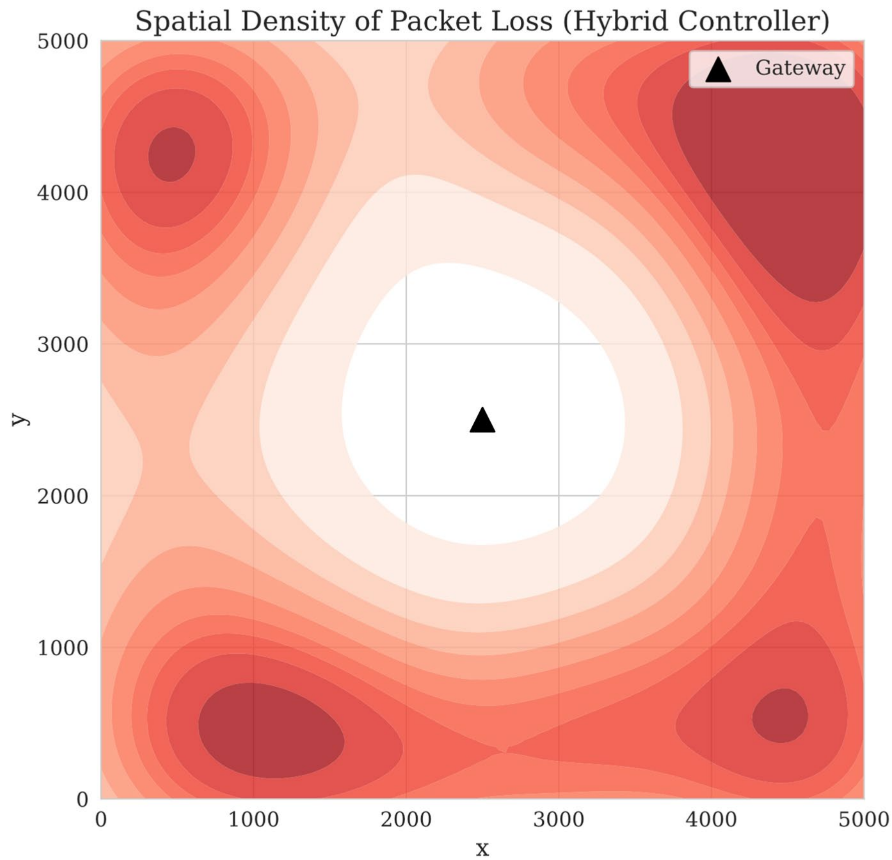

Fig. 8 Spatial density of packet losses. Losses are confined strictly to the cell corners

Hybrid strategy could theoretically extend its operational life by approximately 1.3 years, assuming transmission dominates the energy budget. This gain is achieved without a catastrophic loss in connectivity, as the 87.4% PDR is sufficient for many non-critical monitoring applications, such as asset tracking or smart waste management.

The catastrophic failure of the Standard ADR (46.8% PDR) can be attributed to its Greedy nature, which is fundamentally incompatible with mobility. In a mobile environment, a node moving towards the gateway experiences a rapid, but often transient, improvement in SNR. The Standard ADR interprets this as a stable state and aggressively lowers the Spreading Factor to improve efficiency. However, as the node continues its trajectory, the channel degrades just as quickly. The Standard ADR, with its slow 20-packet averaging window, fails to react to this degradation in time, resulting in a Death Spiral of packet losses. Our Hybrid controller solves this by being Asymmetric: it requires strong, repeated confirmation (4 consecutive checks) to lower the SF for efficiency, but reacts almost immediately (4 packets) to raise power when losses occur for recovery.

The Speed Sensitivity Analysis provides a crucial insight: Standard ADR is not just inefficient; it is functionally broken for vehicular mobility. At 25 m/s, a node traverses 1.5 km in a single minute. A channel state that changes on a minute-by-minute basis cannot be managed by an algorithm based on a 20-minute average. The Hybrid controller's resilience at high speeds suggests that ADR for mobility must be event-driven (loss-based) rather than average-driven. This finding has broad implications for the design of next-generation LPWAN protocols intended for logistics, connected vehicles, and drone tracking.

Conclusions

This study presented the design and rigorous evaluation of a Final Hybrid ADR controller engineered for mobile LoRaWAN environments. By decoupling the logic for rapid loss recovery from that of patient efficiency seeking, the proposed algorithm addresses the fundamental volatility of mobile channels. Using a physics-based Discrete-Event Simulator and extensive Monte Carlo experiments, we benchmarked the proposed solution against Standard (Semtech) and Conservative baselines. Our findings demonstrate that the Hybrid controller achieves superior reliability with a PDR of 87.4%, outperforming the Standard ADR by over 40% points. Furthermore, it reduces energy consumption by 27.6% compared to a conservative baseline, significantly extending device battery life. The controller's robustness is confirmed by its sustained high performance even at vehicular speeds (25 m/s), a scenario where standard ADR performance collapses.

While these results are promising, the study has limitations. The simulation modeled a single-gateway environment. In real-world multi-gateway deployments, macro-diversity could mitigate some mobility effects. Additionally, the standard Log-Distance path loss model used does not account for complex, fast-fading phenomena found in dense urban canyons. Finally, this work establishes the Hybrid ADR's performance against fundamental baselines. Building on this foundation, future research should include a direct, quantitative, simulation-based benchmarking against other advanced mobility-aware schemes, such as PF-ADR, to further contextualize its performance within the field.

Future research will focus on two primary directions. First, we aim to extend the Hybrid logic to Multi-Gateway scenarios, where the network server must aggregate data from multiple reception points. Second, we will explore Machine Learning (ML) approaches. The Smart Switching behavior observed in our rule-based controller suggests that a Reinforcement Learning (RL) agent could learn an even more optimal policy.

Abbreviations

PL	Path Loss
dB	Decibel
d	Distance
L_0	Reference Path Loss
d_0	Reference Distance
γ	Path Loss Exponent
RSSI	Received Signal Strength Indicator
SNR	Signal-to-Noise Ratio
SF	Spreading Factor
LoRa	Long Range
LoRaWAN	Long Range Wide Area Network
ADR	Adaptive Data Rate
RWP	Random Waypoint
Tx Power	Transmission Power
PDR	Packet Delivery Ratio
KPI	Key Performance Indicator
ToA	Time-on-Air
mJ	milliJoule
mW	milliWatt
ID	Identifier

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Authors' contributions
YS performed Data collection, simulation and analysis. LJ evaluate the first draft of the manuscript, editing and writing.

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Data availability
The source code, simulation data, and materials that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Competing interests
The authors declare no competing interests.

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